On application of the problem-solution approach to claims comprising technical and non-technical aspects (especially in the case of computer-implemented inventions), see in this chapter I.D.9.2., and for its application to claims directed to chemical inventions, see in this chapter I.D.9.9.1.

The boards frequently cite R. 42(1)(c) EPC as the basis for the problem- solution approach. R. 42(1)(c) EPC requires that the invention be disclosed in such terms that the technical problem (even if not expressly stated as such) and its solution can be understood. Problem and solution are thus component parts of any technical invention. The problem- solution approach was primarily developed to ensure objective assessment of inventive step and avoid ex post facto analysis of the prior art. As long ago as **T 26/81** (OJ 1982, 211), R. 27(1)(d) EPC 1973 (in the version in force until 31 May 1991; the predecessor of current R. 42(1)(c) EPC) was recognised as clearly binding.

A solution claimed as non-obvious is patentable **only if** it actually has the alleged effect. According to **T 2001/12**, a doubt that the invention as claimed is capable of solving the problem defined in the application may have the following consequences: a) If the claim fails to specify those features which are disclosed in the application as providing the solution to the problem, then an objection under Art. 84 EPC 1973 may properly arise that the claims do not contain all the essential features necessary to specify the invention; b) if this is not the case, but, having regard to the prior art, and irrespective of what may be asserted in the description, it does not appear credible that the invention as claimed would actually be capable of solving the problem, then an objection under Art. 56 EPC 1973 may be raised, possibly requiring a reformulation of the problem. See also **T 862/11** which also dealt with the distinction between the requirements of sufficiency of disclosure (Art. 83 EPC), clarity of the claims (Art. 84 EPC), and inventive step (Art. 56 EPC).

Very early in its case law, the boards emphasised the obligation of objectivity when assessing inventive step; it is the objective rather than subjective achievement of the inventor which has to be assessed (see **T 1/80**, OJ 1981, 206; **T 20/81**, OJ 1982, 217; **T 24/81**, OJ 1983, 133; **T 248/85**, OJ 1986, 261). Starting out from the objectively prevailing state of the art, the technical problem is to be determined on the basis of objective criteria and consideration given to whether or not the disclosed solution is obvious to the skilled person. Although the problem and solution approach is not mandatory, its correct application facilitates the objective assessment of inventive step. The correct use of the problem and solution approach rules out an ex post facto analysis which inadmissibly makes use of knowledge of the invention (**T 564/89**, **T 645/92**, **T 795/93**, **T 730/96**, **T 631/00**). In principle, therefore, the problem and solution approach is to be used; however, if exceptionally some other method is adopted, the reasons for departing from this generally approved approach should be stated.

In **T 967/97** the board stated that the problem and solution approach was essentially based on actual knowledge of technical problems and ways to solve them technically that the skilled person would, at the priority date, be expected to possess objectively, i.e. without being aware of the patent application and the invention that it concerned (see also **T 970/00**, **T 172/03**).

In T 2517/11 the board held that the fact that a technical feature of a known method was "hidden" – i.e. implicit in a prior-art document and not identifiable on a mere reading of that document – and could be detected only by way of a mathematical analysis did not mean that it could not be taken into account as a disclosed feature. If an analysis revealed such a "hidden" feature, that showed it was publicly available; whether there had been any objective reason to carry out the analysis was irrelevant (with reference to G 1/92, OJ 1993, 277). This followed from the objective nature of the problem and solution approach developed in the boards' case law, which entailed consideration of all technical features comprised in the closest prior art, regardless of whether they were directly identifiable or hidden, since even hidden features were publicly available.

The board in T 1761/12 held that the problem and solution approach involved analysing the steps the skilled person would have taken to solve the predefined objective technical problem, and nothing else. Any further reflection on whether the associated changes to the closest prior art identified in this analysis made sense had the effect, in practice, of adding the related aspects of other problems to the objective technical problem initially defined.

In T 320/15 the board held that the problem and solution approach did not consist of a forum in which the appellant (opponent) could freely develop various attacks based on diverse prior-art documents in the hope that one of them would succeed.

Instructive summaries of the case law on the problem and solution approach can be found in a number of decisions; see e.g. R 9/14, T 519/07, T 698/10.

According to the board in T 270/11, the problem and solution approach does not require that the application specify what feature is responsible for producing precisely what advantage or technical effect. All that is required for inventive step is that the claimed subject-matter is not obvious to the skilled person in the light of the prior art (Art. 56 EPC). It is common practice to take features from dependent claims or the description and insert them into an independent claim with a view to rendering the subject-matter patentable and to cite the effects and advantages associated with those features as a basis for (re-)formulating the technical problem. To determine the objective technical problem, the technical results and effects achieved by the claimed invention as compared with the closest prior art must be assessed.

In T 465/92 (OJ 1996, 32) the board did not take the problem and solution approach when assessing inventive step, and said this was merely one possible approach, with advantages and drawbacks. It took the view that all of the seven relevant citations came equally close to the invention. See also T 967/97 in the chapter I.D.3.1. "Determination of closest prior art in general".

In T 188/09 the board noted first that whatever approach was applied as an auxiliary means for the evaluation of inventive step of claimed subject-matter, in a given evidential situation it had to provide the same result, be it either in favour of or against inventive step. Therefore, even if the problem and solution approach was applied, the decision on inventiveness should be the same as if it had not been used. Citing T 465/92

(OJ 1996, 32), the board observed: "if an invention breaks new ground it may suffice to say that there is no close prior art rather than constructing a problem based on what is tenuously regarded as the closest prior art."

In case **R 5/13** (as well as **R 9/13**, **R 10/13**, **R 11/13**, **R 12/13** and **R 13/13** which were all directed against **T 1760/11** of 16 November 2012), the petitioners argued that they should have been allowed to discuss all the issues of inventive step of any stage of the problem and solution approach in respect of all possible starting points that they wished to rely on, despite the fact that the board had structured the discussion by first establishing which document or documents constituted the most promising starting point. The Enlarged Board in **R 5/13** held that the board had not only followed the sequence for the debate announced in its communication annexed to the summons to oral proceedings, but by doing so it had also systematically applied the standard method of the problem and solution approach. The examination whether or not the subject-matter of a patent claim involved an inventive step according to the well-established problem and solution approach was a matter of substantive law. That was equally true for the determination of the closest prior art as the first step of the problem and solution approach, whether one document alone or a plurality of documents was taken as the starting point or most promising springboard aiming at the invention.

3. Closest prior art

3.1. Determination of closest prior art in general

The boards have repeatedly pointed out that the closest prior art for assessing inventive step is normally a prior art document disclosing subject-matter conceived for the same purpose or aiming at the same objective as the claimed invention and having the most relevant technical features in common, i.e. requiring the minimum of structural modifications (see in this chapter I.D.3.2.). A further criterion for the selection of the most promising starting point is the similarity of the technical problem (see in this chapter I.D.3.3.). In a number of decisions, the boards have explained how to ascertain the closest prior art constituting the easiest route for the skilled person to arrive at the claimed solution or the most promising starting point for an obvious development leading to the claimed invention (see in this chapter I.D.3.4. and I.D.3.5.).

In **T 1212/01** the board held that the determination of the closest prior art was an **objective and not a subjective exercise**. It was made on the basis of the notional skilled man's objective comparison of the subject-matter, objectives and features of the various items of prior art leading to the identification of one such item as the closest.

The closest prior art must be assessed from the skilled person's point of view on the day before the filing or priority date valid for the claimed invention (**T 24/81**, OJ 1983, 133; **T 772/94**; **T 971/95**; see also Guidelines G-VII, 5.1 – March 2022 version).

As closest prior art, a "bridgehead" position should be selected, which the skilled person would have realistically taken under the circumstances of the claimed invention. Among these circumstances, aspects such as the designation of the subject-matter of the invention, the formulation of the original problem and the intended use and the effects to be obtained should generally be given more weight than the maximum number of identical technical features (**T 870/96**; see also **T 66/97**, **T 314/15**).

In **T 1450/16** the board did not agree with the assumption relied on by the appellant that, according to the basic problem-solution approach, the person skilled in the art may be entrusted with the task of selecting the closest prior art or a suitable starting point for the assessment of inventive step. It expressed the view that the person skilled in the art within the meaning of Art. 56 EPC entered the stage only when the objective technical problem had been formulated in view of the selected "closest prior art". Only then could the notional skilled person's relevant technical field and its extent be appropriately defined (see in this chapter I.D.8.1). Therefore, it could not be the "skilled person" who selected the closest prior art in the first step of the problem-solution approach. Rather, this selection was to be made by the relevant deciding body, on the basis of the established criteria, in order to avoid any hindsight analysis.

In **T 1742/12** the board endorsed **T 967/97** and **T 21/08**, in which it was found that if the skilled person had a choice of several workable routes, i.e. routes starting from different documents, which might lead to the invention, the rationale of the problem and solution approach required that the invention be assessed relative to all these possible routes, before an inventive step could be acknowledged (see also **T 323/03**, **T 1437/09**, **T 308/09**, **T 1261/14**, **T 1339/14**, **T 259/15**, **T 62/16**, **T 1076/16**). Conversely, if the invention was obvious to the skilled person in respect of at least one of these routes, then an inventive step was lacking (see also **T 558/00**, **T 308/09**, **T 1437/09**, **T 2418/12**, **T 1570/13**, **T 62/16**, **T 64/16**, **T 2443/18**). In **T 967/97** it was further stated that, if an inventive step was to be denied, the choice of starting point needed no specific justification. In **T 261/19** the board stated that this was because the claimed invention must then, as a general rule, be non-obvious having regard to any prior art. The board in **T 1742/12** also held that a piece of prior art may be so remote from the claimed invention, in terms of intended purpose or otherwise, that it could be argued that the skilled person could not conceivably have modified it so as to arrive at the claimed invention. Such prior art might be referred to as "unsuitable". However, this did not prohibit the consideration of an inventive step assessment starting from a piece of prior art with a different purpose (see also **T 855/15**, **T 2304/16**, **T 1294/16**, **T 2443/18**, **T 1112/19**).

If a piece of prior art is **"too remote"** from an invention, it should be possible to show that the invention is not obvious to a skilled person having regard to this piece of prior art (**T 855/15**, **T 2057/12**, **T 2304/16**). A generically different document cannot normally be considered as a realistic starting point for the assessment of inventive step (**T 870/96**, **T 1105/92**, **T 464/98**, **T 2383/17**).

In **T 176/89** the board concluded that the closest prior art comprised two documents in combination with each other. It found that, **exceptionally**, the two documents had to be **read in conjunction**; they had the same patentee, largely the same inventors, and clearly related to the same set of tests. As a rule, however, when assessing inventive step, two documents should not be combined if, in the circumstances, their teaching is clearly contradictory (see also **T 487/95**).

The board in **T 2579/11** ruled that there was no justification for disregarding a priority application and choosing a subsequent application as the closest prior art simply because the description in the subsequent application was by and large more detailed.

Public prior use may be used as the closest state of the art (**T 1464/05**).

In **T 172/03** the board held that the term "state of the art" in Art. 54 EPC 1973 should be understood as "state of technology", and that "everything" in Art. 54(2) EPC 1973 was to be understood as concerning the information which was relevant to a field of technology. It could hardly be assumed that the EPC envisaged the notional person skilled in the (technological) art would take notice of everything in all fields of human culture and regardless of its informational character.

However, the board in **T 2101/12** considered that the interpretation of Art. 54(2) EPC given in **T 172/03** was incorrect. According to the board in **T 2101/12**, the legislator would have used a different term if such meaning had indeed been intended. It held that the wording of Art. 54(2) EPC was clear and required no interpretation. Art. 54(2) EPC itself contains no limitation according to which a non-technical process, such as the signing of a contract at the notary's office, may not be considered state of the art.

In **T 1148/15** the board pointed out that the assumption that the **remaining prior art** was less relevant than the item of prior art identified as the closest may turn out to be wrong, for instance if it could be convincingly shown that the skilled person would have arrived at the claimed subject-matter in an obvious manner when starting from another item of prior art but not when starting from the identified closest prior art. In that situation, or even in case of doubt, the problem-and-solution approach may have to be repeated for any prior art that also qualifies as a suitable starting point.

In **T 405/14** the board stated that the notion of "closest prior art", as it had been developed by the case law of the boards of appeal, appeared to encompass two different meanings, depending on the outcome of the objection raised under Art. 56 EPC. On the one hand, when concluding that a claimed invention was inventive, the notion of "closest prior art" seemed to rely on the assumption that there existed a metric defining the distance between items of prior art and the invention, and that an invention which was not obvious from the "closest prior art" would a fortiori not be obvious with regard to all other items of prior art which, by definition, were not so close. The second meaning was often formulated in terms of a requirement for the "closest prior art" to deal with the same problem as the invention. This was intended to avoid hindsight leading to a finding that inventive step was lacking. The board concluded that here there was no requirement that the "closest prior art" be unique, because the basic rule was that an invention lacked inventive step if it would have been obvious to the skilled person, without hindsight, for any starting point.

In **T 97/14** the board held that a reasoned objection of lack of inventive step must establish the state of the art and set out, in a clear and complete manner, which features of the claimed invention were known from the prior art and where those features could be found in the prior art. In other words, a proper feature mapping was required. This was all the